MIDWEST GEOLOGICAL
SEQUESTRATION CONSORTIUM

What is a Seismic Survey?

Imaging Deep into the Earth with a Seismic Survey

The safe storage of carbon dioxide in rock formations deep in the subsurface depends on proper site selection and care. As part of this process, a seismic survey is used to better understand geological formations in the subsurface before storing carbon dioxide deep underground. This enables imaging of the many layers of rock below the surface by using sound waves. Collection and processing of the data from a seismic survey allows the creation of three-dimensional images of the subsurface. It is a process similar to an ultrasound test that a doctor might perform on a human body, just at a different scale and frequency.

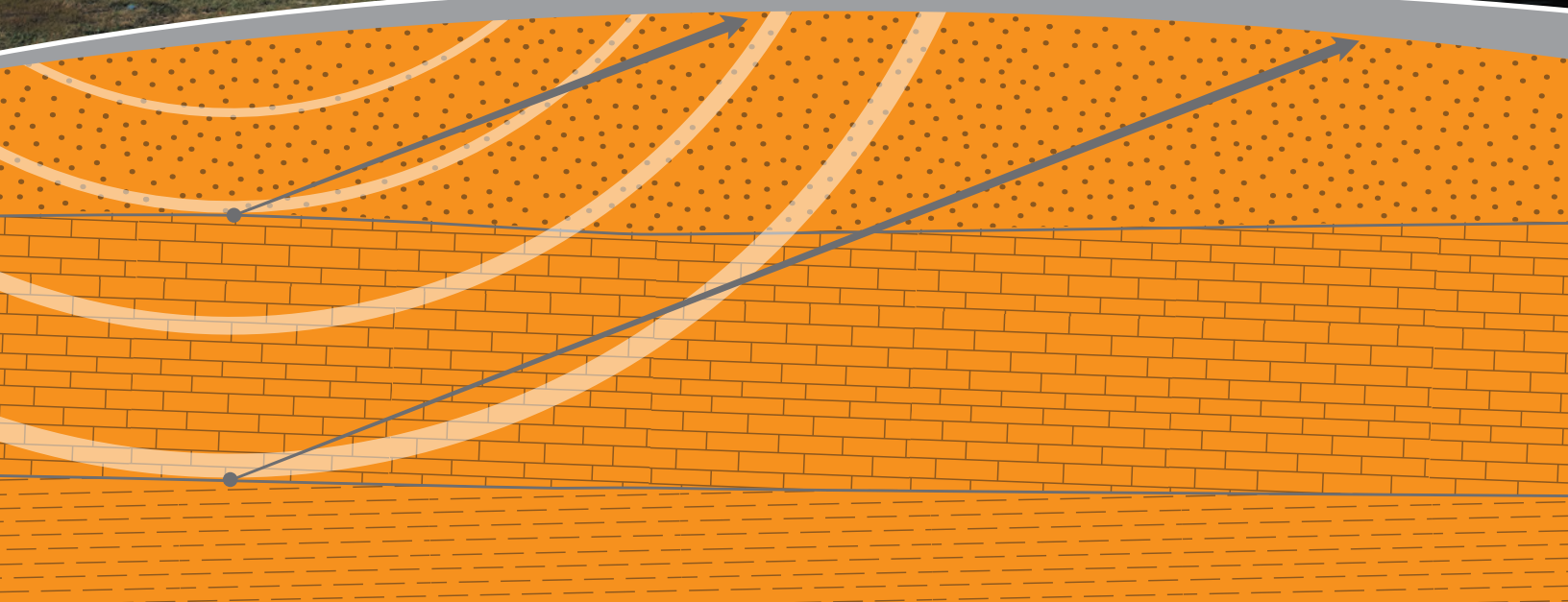
A seismic survey is conducted using receivers or sensors, called geophones, which are installed at the surface to register sound waves reflected from subsurface layers. Geophones consist of a 3-inch above ground portion, which can be seen in the center image and a below-ground spike, which holds the unit securely in the ground. To create the echoes, or waves, special trucks are used. The trucks are equipped with large metal plates called pads that send vibrations through the earth. Notice the vibrator pads located in the center of the vehicle underside in the image on the left.

(*continued on back*)









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First, a survey crew will move down roads or across open fields to gather surface measurements and details. These measurements determine where to place the series of cables equipped with geophones. Once the geophones are placed, trucks move in tandem down the road or through an area lowering a vibration pad onto the surface every few feet. The waves generated travel deep underground and are reflected back as echoes from the different rock layers. The reflections are picked up by a recording truck, which captures the data for computer processing and analysis.

By measuring the time it takes a vibration to reach a rock layer and be reflected back to the surface, the depths of the rock layers can be determined, and an image of the subsurface can be formed. Once the seismic survey data has been collected, any surface sign that the survey has taken place is repaired and within a few short weeks everything is back to the way it was. The subsurface image is created from the data collected which an interpreter then uses to define the shape of the rock layers, how continuous they are, and what type of rock is present. The interpreter also must use data from nearby deep wells to validate the interpretation.

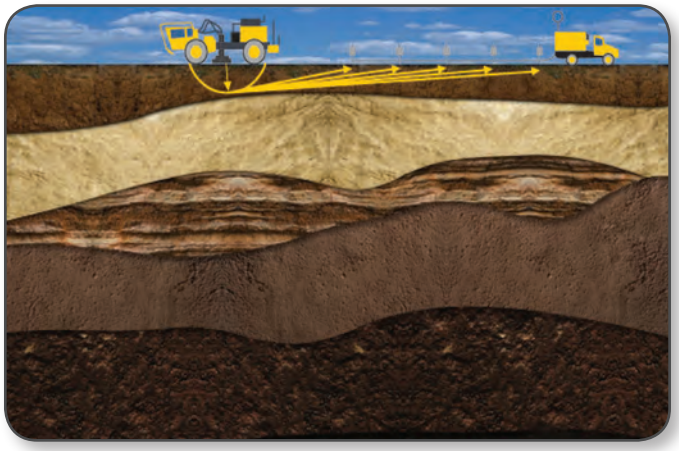


Image Courtesy of Schlumberger

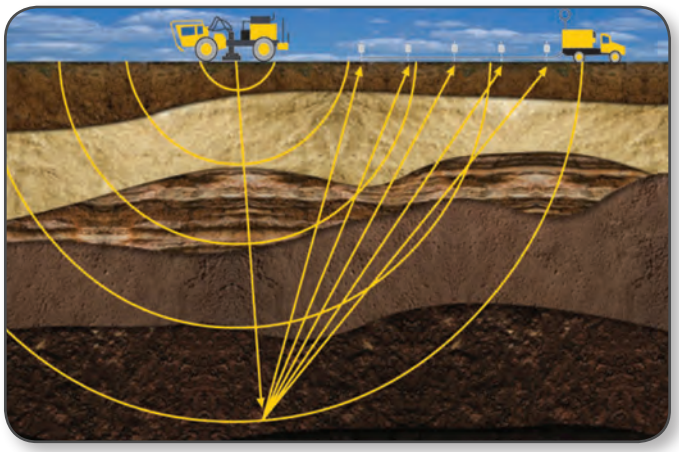
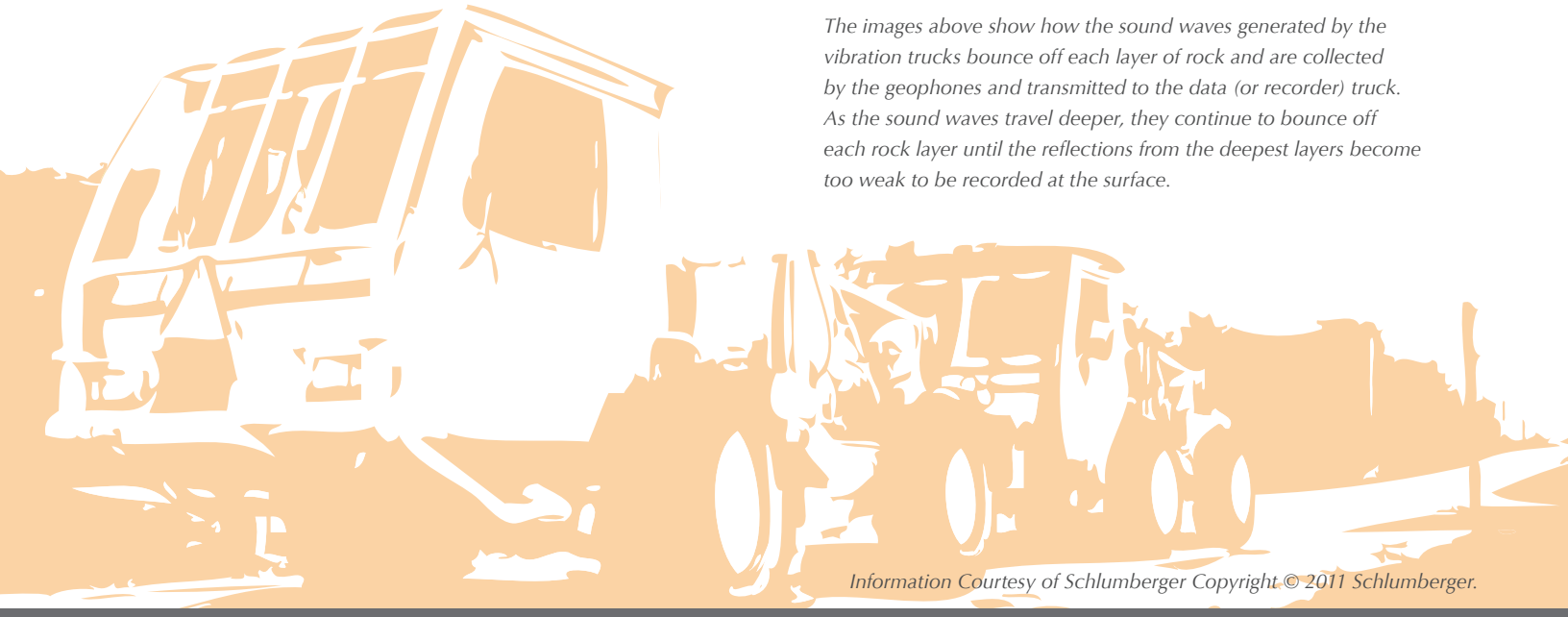


Image Courtesy of Schlumberger

The images above show how the sound waves generated by the vibration trucks bounce off each layer of rock and are collected by the geophones and transmitted to the data (or recorder) truck. As the sound waves travel deeper, they continue to bounce off each rock layer until the reflections from the deepest layers become too weak to be recorded at the surface.



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